

RD-002

AMYOTROPHIC LATERAL SCLEROSIS · LONGITUDINAL PROGNOSTIC MODELLING

Multidomain Longitudinal Prediction of Functional Decline in Amyotrophic Lateral Sclerosis: Cross-Domain and Psychosocial Predictors Beyond Baseline Function

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01 ABSTRACT

Amyotrophic lateral sclerosis (ALS) progresses at markedly different rates across patients and across its major functional domains, including lower-limb, upper-limb, speech, and swallowing function, yet most existing research investigates each domain separately. This investigation asked whether longitudinal, multidomain patient data spanning functional, psychological, behavioural, and care-related measures could predict future ALS functional scores across all four domains within a single modelling framework. Data from 143 ALS patients tracked over multiple visits were tested in 3 models (Persistence Baseline, Elastic Net, and Gradient Boosting) for predicting each patient's walking, hand-use, speech, and swallowing scores 3, 6, and 12 months ahead. Beyond same-domain persistence, current upper-limb function and, in particular, current speech function repeatedly carried predictive information about other domains, most strongly for subsequent swallowing function; hopelessness and somatic depression emerged as the most consistent non-functional predictors across all four outcomes, although their direct bivariate association with subsequent deterioration was weak. Taken together, the results indicate that longitudinal multidomain data are informative about a patient's future functional state, but that predicting the direction and magnitude of individual deterioration remains considerably more limited and requires further validation; the recurring cross-domain and psychosocial signals are therefore best treated as candidate prognostic hypotheses rather than established biological mechanisms or clinically deployable predictors.

02 RESEARCH GAP & OBJECTIVES

Research Gap

- ALS functional decline varies substantially between patients and functional domains, and clinical measures are already used to describe or predict disease progression.
- Little attention has been given to investigations that utilise multidomain longitudinal data to predict future functional status across domains within a single modelling framework, rather than using separate domain-specific analyses.

Research Objectives

- Develop and evaluate machine-learning models predicting future lower-limb, upper-limb, speech, and swallowing ALS functional scores at approximately 3-, 6-, and 12-month horizons from current multidomain data.
- Determine which variables consistently contribute to prediction across domains and horizons by

- It remains unclear whether such predictive information extends meaningfully beyond using a patient's own current functional state to predict future performance.
- More research is needed for examining candidate predictors of deterioration that may establish more prognostic signals for disease severity.
- Much of the existing research investigates prediction of deterioration itself, rather than the patients' future functional levels, which may be more relevant for informing patient-centred care.
- distinguishing same-domain persistence from cross-domain predictive signals.
- Examine whether prominent non-functional predictors are plausibly associated with subsequent deterioration.
- Translate the most robust recurring signals into specific, testable hypotheses for future longitudinal or mechanistic investigation.

03 METHOD

Longitudinal Data Preparation

The source SALSPPD database was restructured into a master longitudinal dataset of approximately 1,001 patient-visit records across 228 variables, with each row representing one patient at one study visit. Database-specific status codes, which represented death, dropout, values not yet measured and other non-numeric states, were converted to missing values before quantitative modelling, since treating them as ordinary measurements would corrupt any model trained on them; clinically meaningful information such as vital status was retained separately rather than discarded outright. Visit-index variables were converted into continuous elapsed time, because raw visit numbers do not correspond to consistent calendar intervals between patients and may lead to misrepresentation of the pace of progression.

Predictive Modelling

For every patient observation, a target future observation was located at approximately 3, 6, and 12 months ahead using each patient's actual visit dates rather than requiring an exact-interval visit, since ALS follow-up intervals are inherently irregular; a tolerance window based on the dataset's observed visit spacing was used to accept the closest available future visit. Nominal categorical variables were one-hot encoded rather than treated as ordered numerical values, to avoid implying a false ranking. 3 modelling approaches were then compared for each of the four functional domains at each horizon: a Persistence Baseline model that carries the current score forward unchanged, an Elastic Net regression, and a Gradient Boosting model. Including the strong “no-change” baseline was essential to establish whether the learned models provided meaningful predictive value beyond the well-documented short-term stability of ALS functional scores.

Validation

All train/test splitting was performed at the patient level. Model performance was assessed using mean absolute error, root-mean-square error, and R^2 for future absolute score, together with a separate direction-accuracy metric: the proportion of cases in which a model correctly identified whether a patient's function would improve, decline, or remain stable. Non-baseline predictor consistency was then assessed by identifying, for each of the twelve outcome $\times$ horizon combinations, which variables ranked among Gradient Boosting's twenty most important predictors, excluding each outcome's own same-domain current score; this isolates genuine cross-domain or non-functional prognostic signal

from the trivial and expected finding that current function predicts near-future function in the same domain. Afterwards, the computational outputs were interpreted and 2 AI-driven literature reviews for candidate biological interpretation were conducted.

Modelling Target

This analysis deliberately modelled future absolute functional score rather than the deterioration increment (current score minus future score) directly. Absolute score was chosen as the primary modelling target because it supports a like-for-like comparison against the persistence baseline on a common, clinically interpretable scale, and because it allows a single framework to be evaluated consistently across all four domains and three horizons. This choice has an important consequence for interpretation, addressed throughout the Results and Findings below: a model can predict future absolute score well largely by reproducing a patient's existing functional level, without necessarily providing useful information about the direction or magnitude of change. Direct change-based modelling is identified as a priority for future work in the Conclusion.

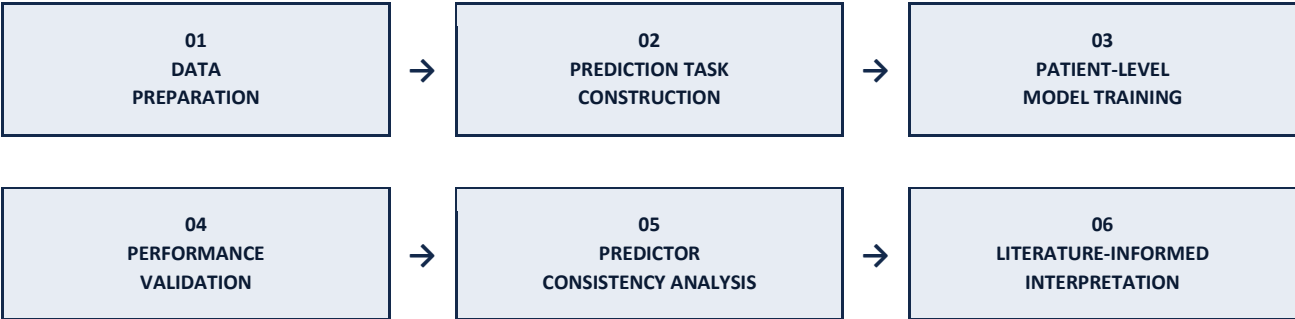
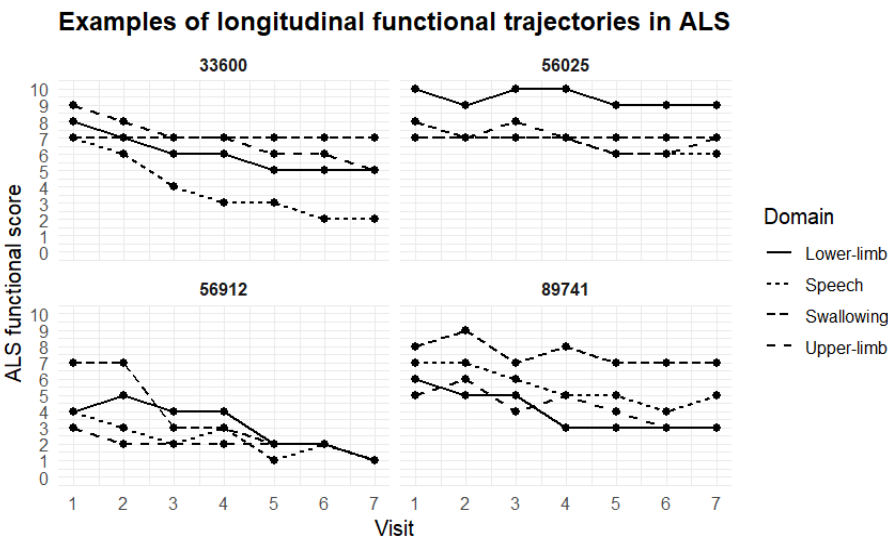


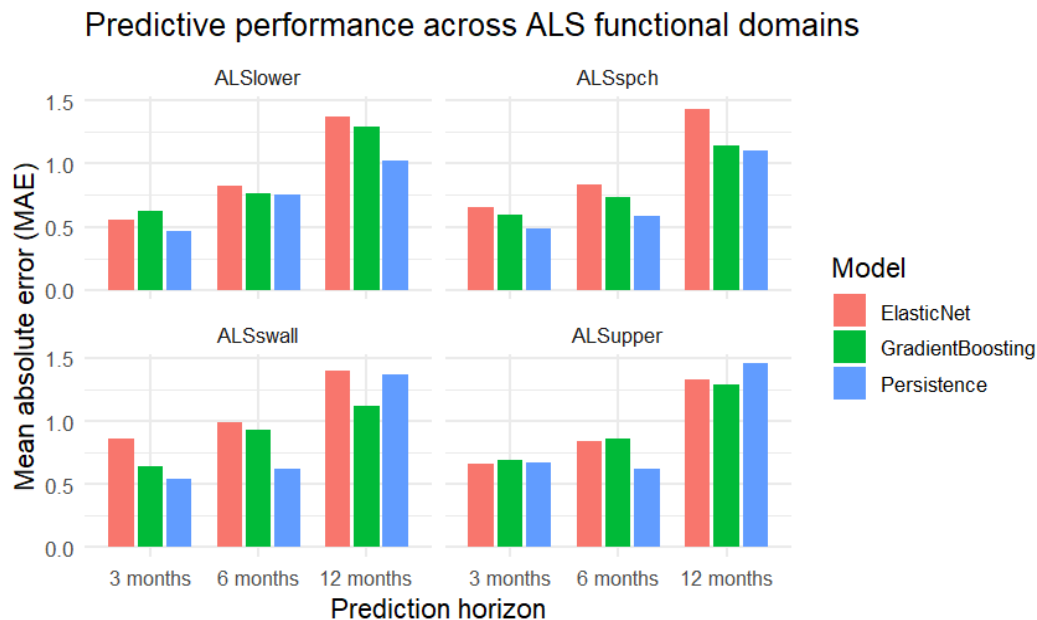
Figure 0. Summary flowchart of the full analytic procedure, from longitudinal data preparation through to literature-informed biological interpretation.

04 RESULTS

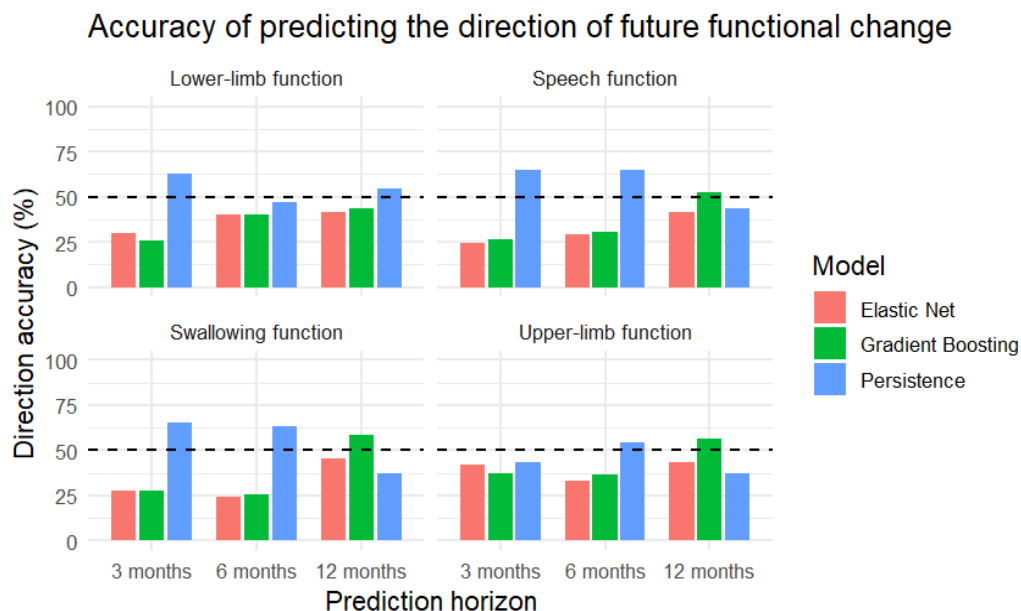
Figure 1. Examples of longitudinal functional trajectories in ALS.



Selected patient trajectories illustrating variation in the timing, magnitude and domain-specific pattern of functional change across lower-limb, upper-limb, speech and swallowing domains. Higher ALS functional scores indicate better function, so downward trajectories represent deterioration. Functional change differs between patients and across domains, with some domains remaining relatively stable while others decline or fluctuate over time.

Figure 2. Predictive performance across ALS functional domains and prediction horizons.

Mean absolute error (MAE) for Persistence, Elastic Net and Gradient Boosting models predicting future lower-limb, upper-limb, speech and swallowing function at approximately 3-, 6- and 12-month horizons. Lower MAE indicates greater accuracy. Increasing horizon was generally associated with greater prediction error, particularly at 12 months. Because ALS functional scores are strongly persistent over time, good prediction of future absolute scores does not necessarily indicate accurate prediction of individual deterioration.

Figure 3. Accuracy of predicting the direction of future functional change.

Percentage of patients for whom each model correctly predicted whether the future ALS functional score would be lower or higher than the current score, at approximately 3-, 6- and 12-month horizons. The dashed line marks the 50% reference level. Despite relatively strong prediction of future absolute scores, machine-learning models showed substantially lower direction accuracy at several shorter horizons — prediction of future state and prediction of individual deterioration are distinct tasks. The apparent improvement in Gradient Boosting direction accuracy at 12 months should be considered preliminary, as only 17 patients contributed to that test set.

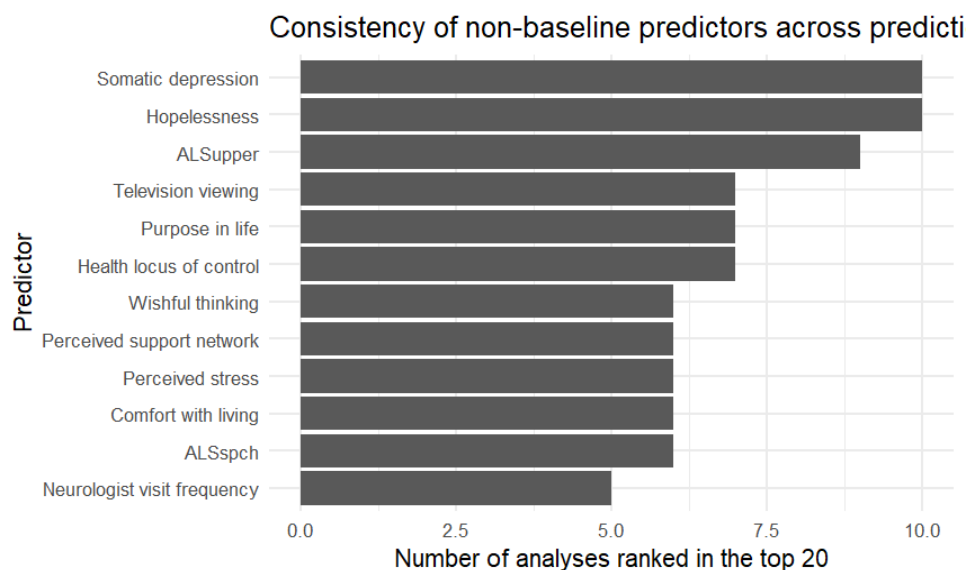
Across the twelve outcome × horizon combinations, Gradient Boosting achieved R^2 values of approximately 0.50–0.94 for future absolute score and improved on the Persistence baseline's R^2 in 11 out of 12 combinations. On mean absolute error (MAE), however, Persistence was frequently competitive with or better than the learned models, particularly at 3 and 6 months. This pattern reverses for direction accuracy at the longest horizon: Gradient Boosting exceeded Persistence on direction accuracy for upper-limb (+19.5 percentage points), swallowing (+21.7 points) and speech (+8.7 points) function at approximately 12 months, while Persistence remained ahead for lower-limb function (−10.8 points). The table below summarises these patterns as domain-averaged values by horizon.

Horizon	Persistence MAE	GB MAE	Persistence R^2	GB R^2	Persistence dir. acc.	GB dir. acc.
3 months	0.54	0.64	0.84	0.86	59.0%	29.4%
6 months	0.65	0.82	0.83	0.86	57.3%	33.2%
12 months	1.24	1.21	0.32	0.51	42.9%	52.7%

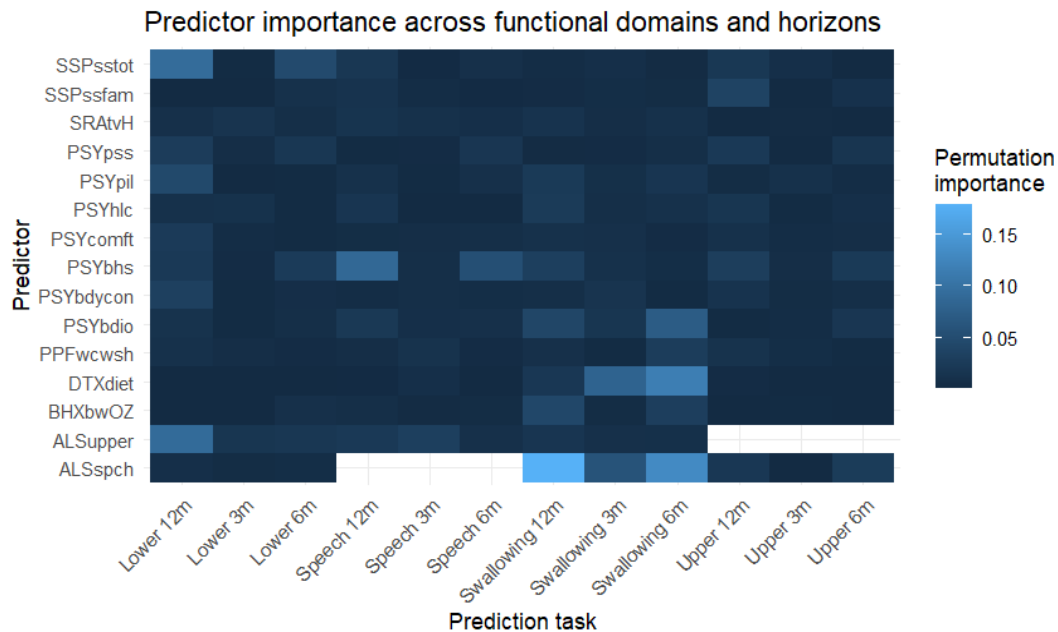
Table 1. Domain-averaged predictive performance by horizon (Gradient Boosting, GB, versus the Persistence baseline).

Same-domain current score was consistently among the strongest predictors identified at every horizon (Section 4 below). Thus, a substantial share of any model's apparent success at predicting future absolute score is attributable to the well-documented short-term persistence of ALS functional status, rather than to genuinely new prognostic information. A model that predicts future score accurately does not, by itself, demonstrate that it has captured clinically meaningful information about progression. Thus, the Persistence baseline is utilised as a mandatory comparison in this study.

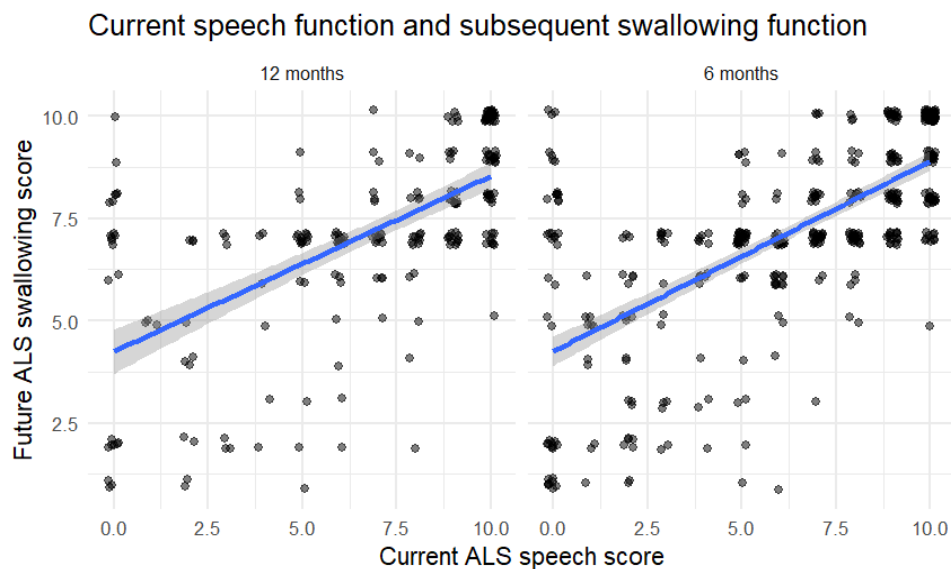
Figure 4. Consistency of non-baseline predictors across prediction tasks.



Number of outcome–horizon analyses (out of a maximum of 12, or 9 for the other ALS functional subscores) in which each predictor ranked among the 20 most important predictors in the Gradient Boosting models. Same-domain baseline ALS scores were excluded because their predictive contribution primarily reflects persistence of existing functional status rather than a novel prognostic signal. Psychological variables, particularly hopelessness and somatic depression, appeared repeatedly across multiple outcomes and horizons.

Figure 5. Distribution of predictor importance across outcomes and prediction horizons.

Permutation importance of the most consistently recurring predictors across the 12 outcome–horizon combinations evaluated using Gradient Boosting. Darker cells indicate greater contribution to predictive performance within the corresponding model. The heatmap illustrates both the recurrence and the context-dependence of candidate predictors across functional domains and horizons; predictor importance represents contribution to model prediction rather than an estimated causal effect.

Figure 6. Association between current speech function and subsequent swallowing function.

Patient-level relationship between current ALS speech score and future ALS swallowing score at approximately 6- and 12-month horizons. Higher scores represent better function. Speech function was repeatedly identified as an important cross-domain predictor of subsequent swallowing function, with its contribution increasing at longer horizons.

Excluding same-domain baseline predictors, hopelessness and somatic depression were the two most consistent predictors overall, each ranking among the top 20 Gradient Boosting predictors in 10 of 12 possible analyses (Figure 4; Figure 5). Current upper-limb function was the most consistent cross-

domain functional predictor, appearing in all 9 analyses for which it could act as a non-baseline predictor, and its association with other domains strengthened at longer horizons. The single strongest specific cross-domain relationship was current speech function predicting future swallowing function (Figure 6), with Gradient Boosting importance rising from approximately 0.13 at 6 months to 0.18 at 12 months, and the corresponding Elastic Net coefficient rising from approximately 0.19 at 3 months to 0.32 at 12 months. Diet texture, television-viewing hours, perceived social-support network size, and neurologist-visit frequency also recurred, generally with more modest and more outcome-specific contributions. Purpose in Life and Health Locus of Control, two further psychological measures other than hopelessness and depression, also ranked among the recurring non-baseline predictors in 7 of 12 analyses each, reinforcing that psychological state as a whole repeatedly carried predictive information.

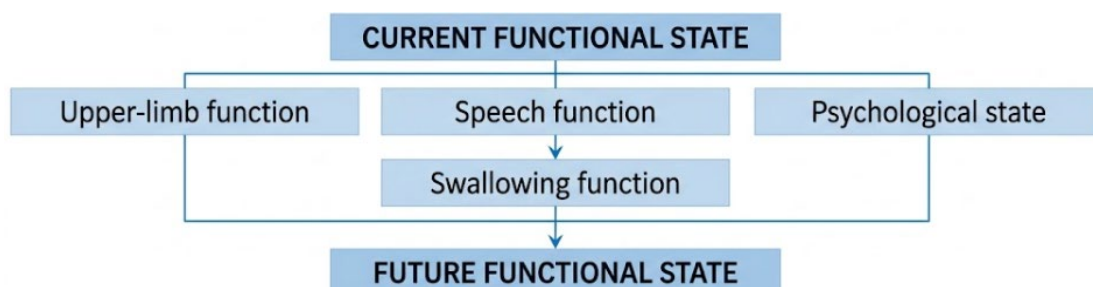
05 FINDINGS

FINDING 01 — FUNCTIONAL DOMAINS PREDICT ONE ANOTHER, ESPECIALLY BULBAR FUNCTION

Once same-domain persistence is set aside, the most consistent predictive signal in the entire analysis was cross-domain: current upper-limb function repeatedly carried information about future lower-limb, speech, and swallowing status, and current speech function was a particularly strong predictor of future swallowing function. Both relationships strengthened at longer horizons. This is consistent with amyotrophic lateral sclerosis behaving as a disease that, while often beginning focally, spreads along anatomically and functionally related pathways over time [1–4,7]; regional progression models built directly on ALS Functional Rating Scale subscores show that decline in one region is informative about, and tends to accelerate, decline in others [5]. The particularly tight coupling between speech and swallowing is consistent with the shared bulbar neuroanatomy underlying both functions: both depend on cranial-nerve-innervated oropharyngeal and laryngeal musculature, and instrumented assessment has found the two impairments to be strongly co-associated in practice [18–21]. Because speech and swallowing subscores are both drawn from the same functional-staging instrument, however, part of this association may reflect a shared underlying disease-stage variable rather than a distinct mechanistic link between the two functions [9,12,17]. The cross-domain associations are therefore reported with high confidence as predictive associations, and with only moderate confidence as evidence of coordinated regional progression specifically.

Implication. Functional domains should not be treated as independent when modelling or monitoring ALS progression. Multidomain modelling can surface relationships such as the speech–swallowing pairing that a domain-by-domain analysis would miss entirely. At the same time, this pairing should be investigated as a coordinated disease pattern worth testing longitudinally, not assumed to be a causal pathway in which impairment in one domain directly produces impairment in another.

Figure 7. Conceptual representation of cross-domain and non-functional predictive associations.



The modelling analysis identified information shared across ALS functional domains, with current upper-limb function repeatedly contributing to prediction of other domains and current speech function showing a particularly consistent association with subsequent swallowing function. These relationships may reflect shared underlying disease burden or coordinated progression across functional systems, but the observational design cannot determine whether one domain directly causes deterioration in another.

FINDING 02 — PREDICTING FUTURE STATE AND PREDICTING DETERIORATION ARE DISTINCT TASKS

At approximately 3 and 6 months, the Persistence baseline correctly identified the direction of a patient's future change in 57–59% of cases on average, compared with 29–33% for Gradient Boosting — despite Gradient Boosting frequently achieving a higher R^2 for future absolute score over the same period. This gap demonstrates that strong prediction of where a patient's score will land is not the same as strong prediction of whether that patient is improving, declining, or stable. At approximately 12 months, this pattern partially reverses: Gradient Boosting's direction accuracy rose to 52.7% on average and exceeded Persistence for three of the four domains. This is broadly consistent with prior work showing that models built primarily on baseline characteristics struggle to anticipate functional change, whereas incorporating accumulated longitudinal information becomes more informative over a roughly annual horizon [63,65,67]; naive baselines have also been reported to be competitive with learned models over shorter windows in comparable ALS prediction tasks [66]. Because the 12-month test set in this analysis comprised only 17 patients, this reversal should be treated as a preliminary, hypothesis-generating result rather than a validated modelling improvement.

Implication. This finding provides only partial support for the underlying research question. Longitudinal multidomain data clearly contain useful information about a patient's future functional level, particularly over shorter horizons — but the study does not establish that these models can reliably identify which individual patients will deteriorate over short time periods. Prognostic modelling in ALS should therefore explicitly separate ‘where will the patient probably be’ from ‘how much will the patient deteriorate’, since the first is answered considerably more easily than the second by the models evaluated here, and future work should weight change-based outcomes accordingly (see Conclusion).

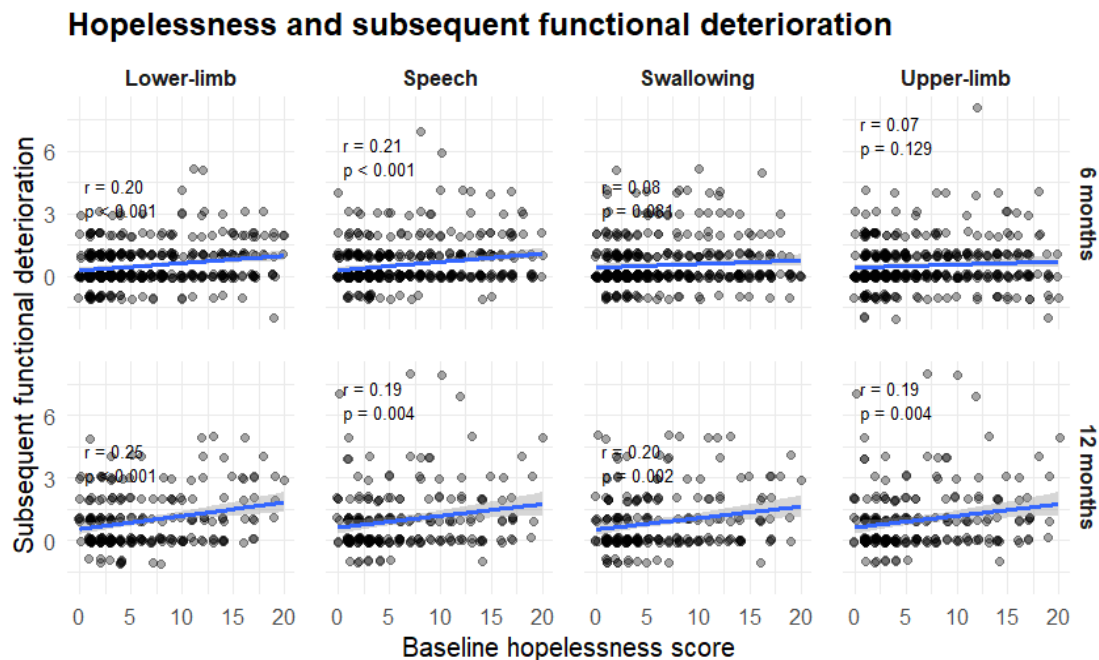
FINDING 03 — PSYCHOLOGICAL DISTRESS, PARTICULARLY HOPELESSNESS, RECURS AS A PREDICTOR BEYOND PHYSICAL FUNCTION

Hopelessness and somatic/organic depression were the two most consistent non-baseline predictors identified anywhere in the analysis, each appearing among the top 20 Gradient Boosting predictors in 10 of 12 possible outcome × horizon combinations, with a consistent direction: greater reported distress was associated with lower predicted future function. Depression, hopelessness, and reduced psychological wellbeing are well documented correlates of ALS disease burden and quality of life [22,23,25,27,28], and hopelessness in particular has been reported to track prospectively with subsequent functional decline independently of baseline disease severity [24]; longitudinal cohort work has similarly found distinct trajectories of worsening psychological symptoms among patients with shorter survival [30]. Because psychological state is measured by a separate self-report instrument rather than derived from physical examination, this association is not simply a restatement of the same functional measurement under another name [23,32]. The somatic depression subscale is a partial exception: symptom inventories that lean on physical items such as fatigue and appetite change are known to be substantially confounded by physical disease burden in ALS populations [33–35], so its predictive value may partly double-count information already present in the physical functional scores.

Hopelessness is accordingly treated with moderate confidence as an independent psychosocial prognostic signal, and somatic depression with lower, exploratory confidence.

Implication. When baseline hopelessness was tested directly against subsequent deterioration (Figure 8), the bivariate association was weak ($r \approx 0.07$ – 0.25) despite hopelessness ranking among the most consistent multivariable predictors anywhere in the analysis. This is an informative finding rather than a contradiction: a variable can contribute substantially to multivariable prediction — through interactions, correlation with other clinical characteristics, or a role as a proxy for broader disease burden — without having a strong simple linear relationship with the outcome on its own. Thus, the results do not validate hopelessness as an independent predictor of ALS deterioration, but rather identify hopelessness as a particularly interesting candidate for dedicated longitudinal follow-up.

Figure 8. Hopelessness and subsequent functional deterioration.



Patient-level relationship between baseline hopelessness score and subsequent functional deterioration (current score minus future score, such that positive values indicate decline) across all four ALS functional domains at approximately 6- and 12-month horizons. Correlations were weak in magnitude ($r \approx 0.07$ – 0.25) but reached statistical significance for most domain $\times$ horizon combinations, and were generally stronger at 12 months than at 6 months, consistent with hopelessness carrying modest but real prognostic information beyond current physical function.

FINDING 04 — DIET TEXTURE IS A CLINICALLY INFORMATIVE BUT LIKELY CIRCULAR PREDICTOR OF SWALLOWING STATUS

Current diet modification level was the strongest non-baseline predictor of near-term (3- and 6-month) swallowing decline, with Elastic Net coefficients of approximately -0.33 and -0.53 respectively. This is unsurprising on clinical grounds: diet texture is modified specifically in response to dysphagia, and documented ALS cohort data show diet adaptation tracking directly with instrumentally assessed swallowing severity [36–39]. Because diet texture is therefore a management response to existing dysphagia rather than an independent measurement of it, its predictive value for future swallowing score most plausibly reflects current swallowing impairment that is not fully captured by the ALS swallowing subscore itself, rather than a distinct prognostic signal. This association is treated as clinically real but not biologically novel.

Implication. Diet texture is best used as a corroborating clinical indicator of current dysphagia severity, not marketed as a newly discovered prognostic marker. Its recurrence in the model is a useful internal sanity check — confirming that the modelling framework recovers clinically expected relationships — rather than a novel scientific contribution in its own right.

FINDING 05 — SOCIAL SUPPORT AND CARE-UTILISATION PATTERNS MAY REFLECT, RATHER THAN PREDICT, DISEASE BURDEN

Perceived social-support network size was associated with lower, not higher, future function across several domains — the opposite of the direction that would be expected if social support were protective. The most plausible explanation is compensatory mobilisation: as ALS progresses and dependency increases, family and caregiver involvement expands, so a larger perceived support network may be a marker of greater care need rather than a protective factor. This interpretation is consistent with caregiving literature showing that caregiver burden and involvement scale with the patient's declining physical function [40–43]. A similar logic applies to neurologist-visit frequency, which also recurred as a predictor: more frequent specialist review plausibly reflects greater clinical concern about a faster-progressing patient (confounding by indication) rather than a treatment effect on subsequent progression [58–60]; observational associations between healthcare or medication exposure and ALS outcomes are well known to be vulnerable to exactly this kind of confounding [58,61,62]. Television-viewing hours followed a comparable pattern and are most plausibly read as a passive behavioural proxy for reduced mobility rather than an independent influence on disease course. All three variables are treated as exploratory prognostic indicators rather than causal factors.

Implication. Behavioural and care-utilisation variables illustrate a general caution for this style of analysis: predictive importance in a multivariable model is not evidence of a protective or harmful effect in either direction. Before any of these three variables is proposed as a monitoring target, its association with deterioration would need to be tested prospectively, with explicit attention to the timing of support mobilisation relative to functional change.

FINDING 06 — HETEROGENEOUS INDIVIDUAL TRAJECTORIES SUPPORT MULTIDOMAIN, RATHER THAN UNIFORM, MODELLING OF PROGRESSION

The individual trajectories shown in Figure 1 (Results) illustrate that ALS functional change is not uniformly smooth, simultaneous across domains, or identical between patients: some patients show relatively stable function across all four domains over the observed visits, others show progressive decline concentrated in a single domain while other domains remain comparatively preserved, and others still show temporary fluctuation rather than monotonic decline. This heterogeneity is precisely what motivates a multidomain, patient-level modelling approach rather than treating ALS progression as a single uniform trajectory summarised by one score.

Implication. This heterogeneity also helps explain why individual deterioration proved difficult to predict at short horizons (Finding 02): two patients with similar current functional scores can plausibly follow different subsequent trajectories, so a model conditioned only on current status has an intrinsic ceiling on how accurately it can anticipate which direction a given patient's course will take. This further supports prioritising direct, change-based and longitudinal statistical modelling approaches in future work, capable of using more than a single current snapshot per prediction.

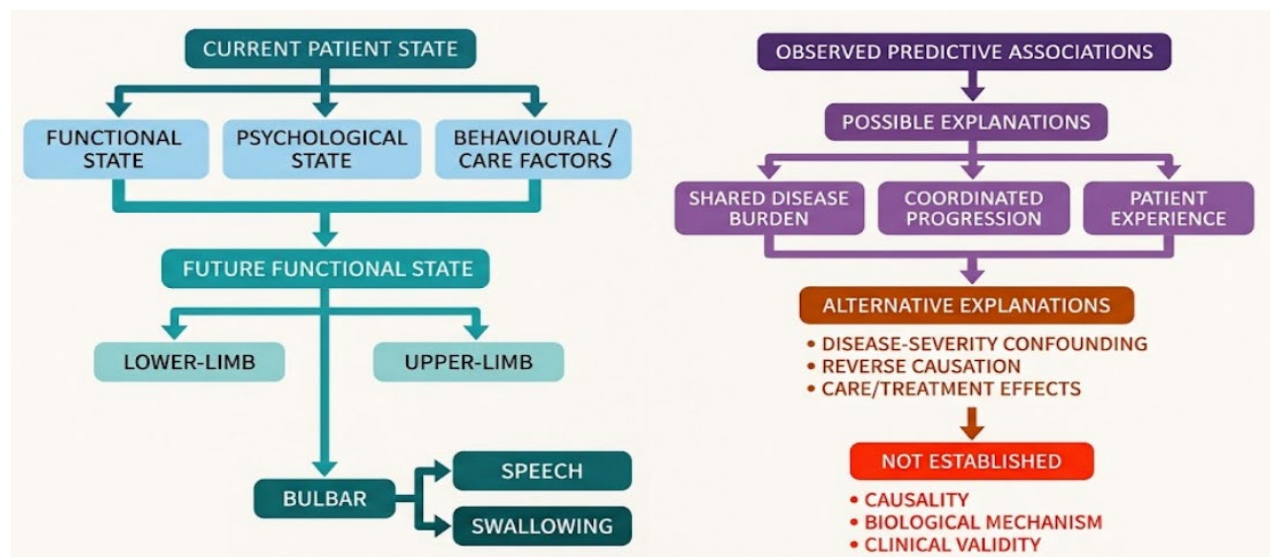
IMPLICATIONS FOR THE RESEARCH QUESTION AND STUDY OBJECTIVES

The findings above provide partial, rather than complete, support for the research question of whether longitudinal multidomain data can predict individual ALS progression. Evidence for objects of the study is outlined below:

1. **Establishing Prediction Models:** Predictive models can be built and compared meaningfully across four functional domains and three horizons, and the mandatory comparison against a persistence baseline shows that a substantial share of predictive performance is attributable to the expected short-term stability of ALS functional status rather than to new prognostic information
2. **Identifying Recurring Predictors:** Several non-baseline predictors, such as hopelessness, somatic depression, Purpose in Life, Health Locus of Control, cross-domain ALSS scores, and selected behavioural and care-utilisation variables, recurred consistently enough across domains and horizons to be considered candidate prognostic signals. However, recurrence in a predictive model is evidence of association, and cannot suggest causal or biological relationship.
3. **Comparing with Associations:** Directly testing the most prominent non-functional predictor, hopelessness, against subsequent deterioration produced only a weak bivariate association. It indicates how multivariable predictive importance and simple, direct association with deterioration are not the same evidence, and that the recurring predictors identified in this study should undergo dedicated, direct testing before being treated as validated prognostic factors.

Implications: Future ALS prognostic tools may benefit from integrating physical functional measures with psychological, behavioural, and care-related information rather than relying on current motor function alone, and, if validated in larger and externally independent cohorts, such multidomain models could eventually help identify patients whose future trajectories warrant closer monitoring.

Figure 9. Framework for interpreting predictive associations across functional and non-functional domains.



The longitudinal modelling analysis identified information shared across functional domains, with current functional status providing substantial information about future functional status. The analysis highlights cross-domain associations and psychological variables in contributing to predictive information. Behavioural, social, and care-utilisation variables may also contain prognostic information, but may themselves be shaped by disease severity or changes in patient circumstances.

06 LIMITATIONS

Several limitations affect how the findings above are discussed below.

Sample size and dimensionality. The analysis is internally large in visit-rows but small in independent patients: 143 patients contributed the underlying data, and only 17 remained in the test set at the 12-month horizon. Against this, 178–201 candidate features were considered at each horizon.

Single holdout split. A single train/test division was used rather than repeated resampling or cross-validation. The strongest claims in this report — same-domain persistence and the upper-limb and speech–swallowing cross-domain associations — were consistent across all three horizons and both models, which offers some internal robustness but have not been confirmed under an independent resampling procedure.

No external validation. All findings are internal to this dataset and this study population — three US hospital-based ALS clinics and affiliated community support groups, with accrual beginning in 1987–88 [50]. Generalisability to contemporary or differently composed ALS populations has not been tested.

Observational design. The data are observational, and every association reported here is a predictive association rather than a demonstrated causal effect. Directionality, particularly for hopelessness, social support, and neurologist-visit frequency, cannot be established from a single cross-sectional predictor measured alongside the outcome.

Approximate horizons. The 3-, 6-, and 12-month horizons are approximate, visit-based intervals rather than exact elapsed calendar time, and should not be read as precise temporal rates of change.

Missingness, dropout, and death. The source database explicitly codes death, dropout, and unmeasured visits, and the wider longitudinal-methods literature establishes that such missingness is rarely unrelated to the underlying trajectory it interrupts [46–49]. Follow-up availability is therefore plausibly non-random with respect to disease severity, and predictors that appear more informative at 12 months could partly reflect which patients remained measurable at that horizon.

Construct overlap. Diet texture and swallowing score, speech and swallowing subscores, and somatic depression items and physical symptom burden are each plausibly measuring overlapping, rather than fully independent, constructs [33–36].

Feature importance is not biological importance. Gradient Boosting permutation importance and Elastic Net coefficients describe each variable's contribution to predictive performance within these specific models. They do not, by themselves, indicate biological importance, causal effect size, or clinical significance.

Outcome definition. The primary modelling target was future absolute functional score rather than the deterioration increment directly. This was a deliberate scope choice to enable a consistent, comparable framework against the Persistence baseline across all domains and horizons. Thus, the study is better positioned to speak to future-state prediction than to deterioration prediction.

Bivariate follow-up testing. Direct testing against subsequent deterioration was carried out for hopelessness only (Finding 03); the other recurring non-baseline predictors identified in Section 4 and Section 5 have not yet been subjected to the same direct check, and their status as candidate prognostic signals should be treated as provisional pending that analysis.

07 AI INTEGRATION

The bars below document which named AI agents contributed to each stage of this investigation, and their relative share of the AI-assisted work within that stage. Bars reflect the proportion of AI usage only, without accounting for human operations and oversight at each stage.

DATA PROCESSING



MATHEMATICAL MODEL BUILDING (PYTHON)



DATA INTERPRETATION



BIOLOGICAL INTERPRETATION



VALIDATION OF CLAIMS



08 CONCLUSION

This investigation asked whether longitudinal, multidomain patient data could predict future ALS functional status across four domains, and which variables beyond current functional state consistently provided prognostic information. The study provides **partial support**. Gradient Boosting models explained substantial variance in future functional scores and generally improved on a persistence baseline, indicating that earlier multidomain information can provide useful information about subsequent functional status. However, direction accuracy was generally weak at 3 and 6 months, meaning the models could not reliably identify which individual patients would deteriorate over short intervals. At approximately 12 months, Gradient Boosting exceeded persistence for upper-limb, speech and swallowing outcomes.

Current functional status was consistently the strongest predictor, reflecting persistence of disease state rather than a novel finding. Beyond this, recurring psychological, behavioural and cross-domain variables — particularly hopelessness, upper-limb function and speech function — provided candidate prognostic signals. Further testing revealed weak direct association between hopelessness and subsequent deterioration, thus demonstrating that predictive importance in a multivariable model does not necessarily indicate a strong direct relationship. Disease severity, reverse causation and construct overlap may also explain some observed associations.

Overall, the findings suggest that multidomain longitudinal data can meaningfully estimate a patient's future functional state, but predicting the direction and magnitude of individual deterioration remains considerably harder. The recurring cross-domain and psychosocial signals therefore represent testable hypotheses that may be investigated in future studies.

Future Directions

Model deterioration directly. Future work should model the deterioration increment (current score minus future score) as the primary target, rather than future absolute score alone, to align the model directly with the clinically relevant outcome and separate it more cleanly from same-domain persistence.

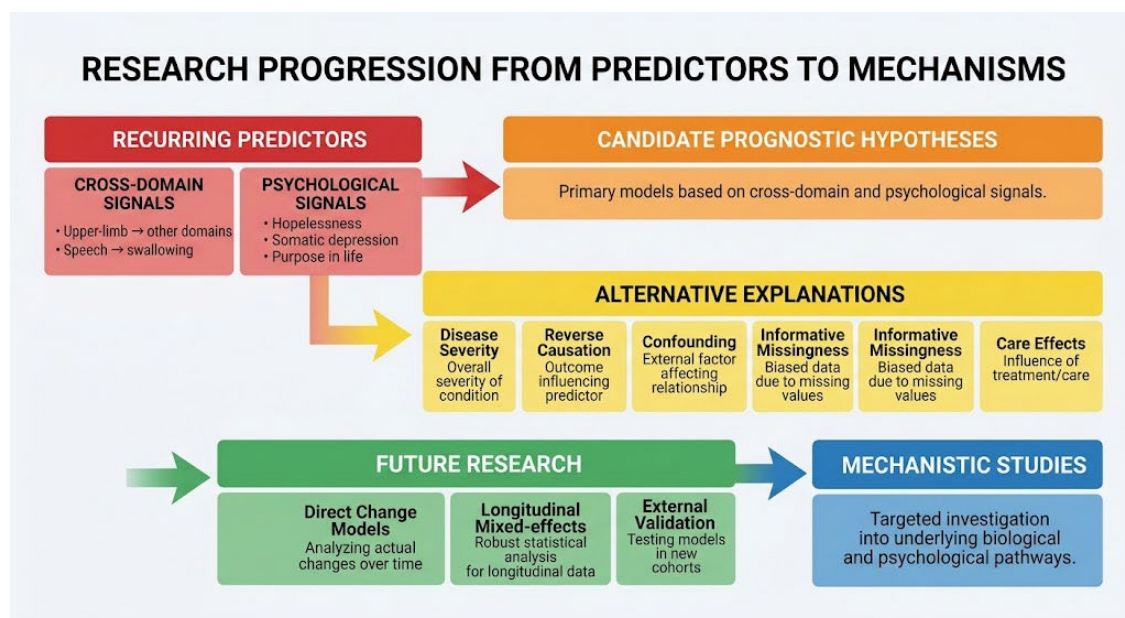
Use longitudinal statistical methods designed for repeated measures. Because patients contribute multiple correlated observations, mixed-effects models, longitudinal regression, repeated-measures approaches, and time-to-event models where appropriate would better account for within-patient correlation than the single-snapshot approach used here.

Externally validate the recurring predictors. Hopelessness, somatic depression, cross-domain ALSS scores, and the behavioural and care-utilisation variables identified here should be tested in larger and independent ALS cohorts; recurrence within this single dataset does not establish that a predictor will generalise elsewhere.

Investigate the psychological predictors more directly. Hopelessness, depression, Purpose in Life, and Perceived Stress warrant dedicated follow-up to determine whether they independently predict progression, reflect disease severity, mediate other relationships, interact with functional impairment, or change alongside disease progression.

Test cross-domain progression directly. The speech–swallowing relationship provides a specific, testable hypothesis: whether longitudinal change in bulbar function shows coordinated progression that improves prediction of subsequent swallowing impairment, evaluated using longitudinal models of change rather than feature importance in a single-snapshot predictive model.

Figure 10. From predictive associations to testable prognostic hypotheses.



The modelling results generate candidate prognostic hypotheses from recurring predictive signals. Cross-domain functional associations and repeatedly identified psychological variables provide potential targets for further investigation, but these associations may alternatively reflect overall disease severity, reverse causation, confounding, informative missingness or dropout, or changes in care. Future studies should evaluate these signals using direct measures of functional change, longitudinal statistical models, larger cohorts, and extensive external validation.

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ALSFRS-R and clinical staging

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